

TF2 Refresh & Static/Dynamic Frame Project Summary

Learning Refresh

I refreshed my knowledge of ROS 2 tf2 and improved my understanding of coordinate frames and spatial relationships in robotics. I learned the difference between static and dynamic transforms and reviewed the basic coordinate-frame tree used in a robot.

1. Key TF2 Concepts

Coordinate frames: A robot is represented using multiple coordinate frames. Each frame describes a position and orientation relative to another frame. tf2 manages these relationships and allows ROS 2 nodes to calculate how one frame is positioned relative to another.

Static transform: Used when the relationship between two frames does not change. For example, a sensor permanently mounted on a robot has a fixed position relative to base_link.

Dynamic transform: Used when the relationship changes over time. For example, a moving part, joint, or simulated frame can continuously change its position or orientation relative to its parent frame.

2. Basic Robot TF Tree

A common robot hierarchy is:

```
map → odom → base_link → camera_link / sensor_link
```

map represents the global/world reference frame, odom represents the robot's odometry frame, and base_link represents the robot body. Sensors and other robot components are normally attached as child frames of the appropriate parent frame.

3. Project Explanation — Static/Dynamic Frame Drill

The project is a small ROS 2 tf2 exercise designed to demonstrate the difference between static and dynamic transforms. In this simplified drill, base_link is the root/reference frame. Two child frames are created:

Frame	Type	Purpose
base_link	Root / reference	Represents the robot body and is used as the reference for the drill.
mock_sensor_link	Static	Represents a fixed sensor mounted on the robot. Its offset from base_link stays constant.
mock_frame	Dynamic	Represents a moving mock frame whose position/orientation changes over time.

The simplified frame tree for the project is:

```
base_link
├─ mock_sensor_link (static — fixed offset, e.g. x=0.20 m, y=0.00 m, z=0.15 m)
```

└─ mock_frame (dynamic — position changes over time)

4. Important Understanding: base_link

In this project, base_link is the root frame. This does not mean that base_link itself is a transform that is being published to something else. It is simply the reference frame from which the child-frame relationships are defined in this simplified exercise.

The important idea is that a TF transform describes the relationship between two frames. For example, base_link → mock_sensor_link tells ROS where the sensor frame is located and oriented relative to base_link.

5. What the Project Demonstrates

- A static transform keeps the sensor at a fixed position relative to base_link.
- A dynamic transform updates as the mock frame moves.
- The TF tree shows how frames are connected through parent-child relationships.
- RViz2 can visualize the frames and their axes so the transforms can be checked visually.
- The same concepts can later be applied to real robot sensors and real robot motion.

6. Short Takeaway

This project is a practical refresh of tf2: understand the frame tree, publish a fixed sensor transform, publish a changing transform for a moving frame, and verify the complete setup in RViz2. The goal is not a complex robot simulation; it is to prove that the coordinate-frame relationships are understood and correctly connected.